

Machine Learning — Course Syllabus

Demo syllabus for Learnova AI testing

Course Duration: 12 Weeks **Level:** Beginner to Intermediate

Prerequisites: Basic Python programming, basic statistics, and familiarity with linear algebra.

Course Objectives

By the end of this course, students should be able to understand core machine learning concepts, prepare datasets, train and evaluate supervised and unsupervised learning models, compare algorithms, and build a complete machine learning workflow.

Unit 1: Introduction to Machine Learning

1. What is Machine Learning?
2. AI vs ML vs Deep Learning
3. Types of Machine Learning
4. Supervised, Unsupervised and Reinforcement Learning
5. Machine Learning Workflow
6. Applications and Real-world Use Cases

Unit 2: Python and Data Preparation

1. NumPy fundamentals
2. Pandas fundamentals
3. DataFrames and data manipulation
4. Data cleaning
5. Handling missing values
6. Categorical encoding
7. Feature scaling

Unit 3: Exploratory Data Analysis

1. Descriptive statistics
2. Data visualization
3. Correlation analysis
4. Outlier detection
5. Feature distributions
6. Train-validation-test split

Unit 4: Supervised Learning — Regression

1. Linear Regression
2. Multiple Linear Regression
3. Polynomial Regression
4. Regularization: Ridge and Lasso
5. Regression metrics: MAE, MSE, RMSE and R^2

Unit 5: Supervised Learning — Classification

1. Logistic Regression
2. K-Nearest Neighbors
3. Decision Trees
4. Random Forest
5. Support Vector Machines
6. Classification metrics
7. Confusion Matrix, Precision, Recall and F1-score

Unit 6: Unsupervised Learning

1. Clustering concepts
2. K-Means Clustering
3. Hierarchical Clustering
4. DBSCAN basics
5. Dimensionality Reduction
6. Principal Component Analysis (PCA)

Unit 7: Model Evaluation and Selection

1. Bias and Variance
2. Overfitting and Underfitting
3. Cross-validation
4. Hyperparameter tuning
5. Grid Search and Random Search
6. Learning curves

Unit 8: Feature Engineering

1. Feature selection

2. Feature extraction
3. Numerical transformations
4. Interaction features
5. Handling imbalanced datasets
6. Data leakage

Unit 9: Ensemble Learning

1. Bagging
2. Boosting
3. AdaBoost
4. Gradient Boosting
5. XGBoost overview
6. Random Forest revisited

Unit 10: Introduction to Neural Networks

1. Neurons and activation functions
2. Perceptron
3. Forward propagation
4. Loss functions
5. Backpropagation concept
6. Introduction to deep learning

Unit 11: Machine Learning Projects

1. Problem definition
2. Dataset selection
3. Data preprocessing
4. Model training
5. Evaluation and comparison
6. Model interpretation
7. Project documentation and GitHub

Unit 12: Deployment and Responsible ML

1. Saving trained models
2. Inference pipeline
3. Basic API deployment

4. Model monitoring concepts
5. Fairness and bias
6. Privacy and responsible AI

Assessment Pattern

Assessment	Weight
Topic Quizzes	15%
Unit Tests	20%
Assignments / Labs	20%
Mid-Term Examination	15%
Final Project	20%
Final Mock Test	10%

Recommended Practical Work

- Build a house-price prediction model using regression.
- Build a classification model for customer churn or disease-risk prediction using an appropriate public dataset.
- Perform customer segmentation using K-Means clustering.
- Create a complete ML pipeline from data preprocessing to model evaluation.
- Document one project with code, results, limitations, and future improvements on GitHub.

Important Topics: Linear Regression, Logistic Regression, Decision Trees, Random Forest, SVM, K-Means, PCA, Cross-Validation, Overfitting/Underfitting, Feature Engineering, Model Evaluation, and ML Project Workflow.

This is a fictional/demo syllabus created for testing the Learnova AI syllabus-upload and analysis workflow.